



# Auto-ignition behaviors of nitromethane in diluted oxygen in a rapid compression machine: Critical conditions for ignition, ignition delay times measurements, and kinetic modeling interpretation

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## ABSTRACT

In this work, the weakest thermodynamic conditions for the auto-ignition of mixtures containing nitromethane were experimentally determined by using the rapid compression machine facility. Results show there is a narrow weak ignition region between ignition and non-ignition. The weak ignition region would disappear with the increase of the EOC (end of compression) pressure and nitromethane concentration. In addition, the ignition delay times for successful auto-ignition for different nitromethane concentrations and equivalence ratio mixtures were measured and compared. Results show that the dependence of nitromethane ignition on the equivalence ratio is weak. Subsequently, the measured ignition delay time data were employed to validate several kinetic models in literature and our previous model shows better agreement with experimental results, as well as other available literature data. Sensitivity analysis for the model reveals the importance of unimolecular decomposition and H-abstraction reactions for the ignition delay times in the temperature range studied herein. Finally, critical conditions for nitromethane ignition under extended conditions that are beyond the ability of the experimental facility were predicted.

## 1. Introduction

Safety testing of energetic materials such as explosives or propellants is highly desirable. To measure thermal properties as time to explosion (ignition delay time) and weakest thermodynamic condition for thermal explosion is very important for the construction of the cook-off models and for the safety issue during the storage, transportation and handling of energetic material [1].

As an energetic material, nitromethane (NM or  $\text{CH}_3\text{NO}_2$ ) is the simplest nitro-paraffin and has been selected as prototype compound to study the combustion behaviors of practical explosives and propellants such as research department explosive (RDX) [2] and hydroxyazanium nitrate (HAN)-based monopropellant [3]. NM has been used as a monopropellant and a liquid explosive. In addition, it can be used as an additive for internal combustion engines, which can increase the power output and reduce soot emission [4–6]. Furthermore, as an explosive compound, accidental leakage of nitromethane into air forms explosive and flammable mixtures [7,8], which leads to serious risk of safety, which are mainly caused by thermal or shock stimulations.

Extensive theoretical and experimental investigations have been conducted to evaluate the explosive hazards based on nitromethane at

different levels [9–16]. The processes of detonation in pure NM have been studied by plate impact experiments under a steady one-dimensional strain for pressures from 8.5 to 12 GPa [13]. The results not only reveal the main steps of the classical homogeneous model, but also show that the build-up process is extremely complex. In addition, the explosion process of mist/air and mist/dust/air of NM has been reported by Liu et al. [17]. For the pure gaseous nitromethane mixtures, cellular structure of detonation [8,18] and the influence of nitromethane concentration on ignition energy [7] in gaseous nitromethane mixtures have been reported. However, the lowest temperature for auto-ignition of a given mixture containing nitromethane has not been reported.

In addition, understanding the auto-ignition behaviors of nitromethane is important for the hazardous evaluation of this energetic material. The auto-ignition characteristics are primarily controlled by the dissociation or oxidation kinetics of nitromethane under different thermodynamics conditions. As such, a number of theoretical and experimental researches [19–27] on the decomposition and oxidation kinetics of nitromethane have been conducted. These work showed that the main nitromethane decomposition paths include the direct C–N bond cleavage to generate  $\text{CH}_3$  and  $\text{NO}_2$ , and O–N bond cleavage to

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generate  $\text{CH}_3\text{O}$  and  $\text{NO}$ . Furthermore, ignition delay times of  $\text{CH}_3\text{NO}_2/\text{O}_2/\text{Ar}$  and  $\text{CH}_3\text{NO}_2/\text{O}_2/\text{N}_2$  mixture have also been measured by Mathieu et al. [25] and Nauc  r et al. [26], for temperatures higher than around 900 K. These two papers both found a two-stage ignition behavior. Meanwhile, Mathieu et al. developed a detailed kinetic mechanism, which showed reasonable performance in predicting their measured ignition delay times at high temperature. Recently, we have also measured the ignition delay times of  $\text{CH}_3\text{NO}_2/\text{O}_2/\text{Ar}$  mixtures at high temperatures (1200–2000 K). Additionally, we developed a detailed kinetic mechanism for nitromethane oxidation through ab-initio calculations, which shows good agreement with our measured ignition delay times as well as the literature data, including laminar burning velocities and ignition delay times at high temperature [27]. However, there has been no report of the experimental and simulated combustion parameters of nitromethane in low to intermediate temperature range.

Since the critical condition that leads to nitromethane auto-ignition provides some guidelines for the safety evaluation of nitromethane, and understanding of the auto-ignition onset is of merit for designing new safety testing methodology for energetic materials. Our first objective of this work is to experimentally determine the critical thermodynamic conditions (lowest temperature) for the auto-ignition of a given mixture containing nitromethane. Secondly, for the successful auto-ignition cases, ignition under a given thermodynamic condition is initiated after a certain period of ignition delay time, which represents the overall reactivity of the mixture, thus we will provide ignition delay data of nitromethane, in the low to intermediate temperature range for different pressures and mixture conditions by using a rapid compression machine facility. The data will then be used to validate our recently proposed kinetic model that includes updated low temperature chemistry, after which sensitivity and reaction pathway analysis based on our validated model will be presented to interpret the ignition kinetics. Finally, by using the validated model, the critical thermodynamic conditions at extended conditions that are beyond the ability of the experimental facility will be predicted.

## 2. Experimental specifications

### 2.1. Setup and procedures

Ignition delay times of nitromethane (Aladdin, 99.0%) were measured using the rapid compression machine facility. Fig. 1 shows the schematic image of the rapid compression machine system. Details of the RCM system have been introduced in our previous work [28]. Briefly, the system consists of six parts, including the high pressure air

tank, the driving chamber, the hydraulic chamber, the compression chamber, the combustion chamber and the control and data acquisition system. The pressure evolution inside the combustion chamber section is recorded by a recess-mounted piezoelectric pressure transducer (Kistler 6125C) combined with a charge amplifier (Kistler 5018A). Details of the operation of the rapid compression machine can be found in the Supplementary material.

The reactive and non-reactive mixtures are prepared in two heated separate stainless vessels according to the partial pressure of each component. A relative manometer (OMEGA DPG4000) with an accuracy of 10 Pa is used to measure the partial pressure of each component. It is noted that the partial pressure of the liquid nitromethane is assured to be less than 1/3 of its saturated pressure at the vessel temperature so as to avoid condensation effect. We note that the saturated vapor pressure of nitromethane at room temperature (293 K) is small (3.71 kPa). The mixture preparation system including the mixture vessel, the pipelines and the reaction chamber is heated with temperature fixed at 328 K. At this temperature, the vapor pressure is 20 kPa [29]. The maximum partial pressure of nitromethane in the experiment is assured to be less than 7 kPa. 8 mixtures are prepared and the components of each tested mixture are summarized in Table 1. In these mixtures, Mix 1–4 and Mix 8 at 15 bar are used to study the critical conditions for nitromethane ignition; While Mix 5–8 is used to study the chemical kinetic of nitromethane. Because the nitromethane concentration is low in this experiment, the explosion pressure after ignition is expected to be low and thus the experiment is safe. In addition, to find the appropriate diluents composition for reliable ignition delay time determination in rapid compression machine (typically 10–200 ms), several tests have been performed. When using Ar as the diluents, ignition happens very fast and for some cases ignition happens even before EOC. When  $\text{N}_2$  is used as diluents, ignition is not observed within 200 ms. The reason is that nitrogen has large specific heat, which leads to low temperature after compression. As such, the  $\text{Ar}/\text{N}_2$  ratio is then fixed at 3:1. Furthermore, the highest temperature in the present study is limited by the shortest ignition delay times that can be validly extracted from the pressure trace. The corresponding non-reactive mixture experiments were conducted by replace  $\text{O}_2$  using the same mole fraction of  $\text{N}_2$  to obtain volume-time histories.

### 2.2. Definition of ignition delay times

Fig. 2 shows a typical pressure evolution history for the ignition of  $\text{CH}_3\text{NO}_2/\text{O}_2/\text{N}_2/\text{Ar}$  mixture (solid line) and the corresponding non-reactive mixture (dash line). The ignition delay in this work is defined as

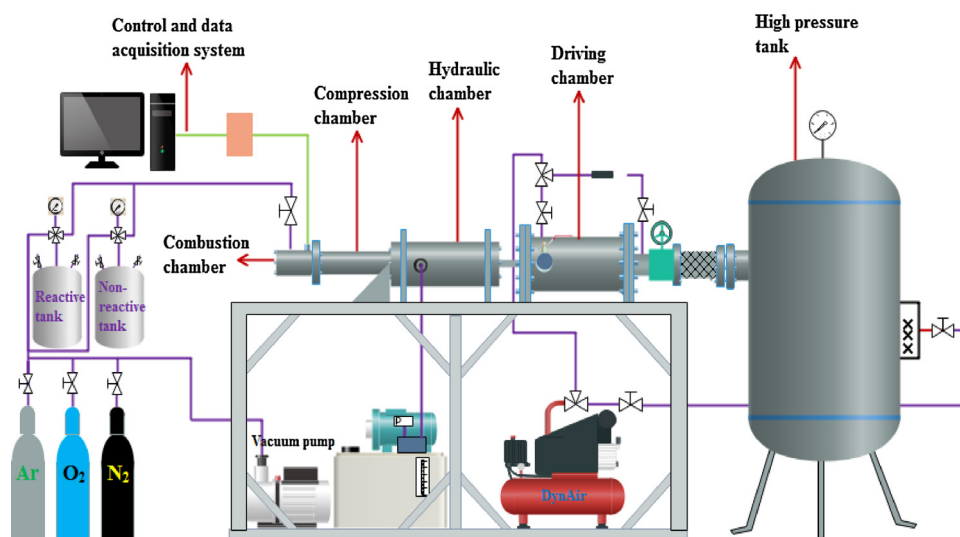


Fig. 1. Schematic of the rapid compression machine system for the present ignition delay time measurements.

**Table 1**  
Test conditions and reactant mixtures.

| Mix # | $\phi$ $P_c$ Ar/N <sub>2</sub> |        |   | Mole fraction (%) |                   |                   |
|-------|--------------------------------|--------|---|-------------------|-------------------|-------------------|
|       |                                |        |   | [NM]              | [O <sub>2</sub> ] | [N <sub>2</sub> ] |
| 1     | 2.0                            | 10–30  | 3 | 1.00              | 0.625             | 24.59             |
| 2     | 2.0                            | 15     | 3 | 0.5               | 0.3125            | 24.80             |
| 3     | 2.0                            | 15     | 3 | 1.5               | 0.9375            | 24.39             |
| 4     | 2.0                            | 15     | 3 | 2.5               | 1.5625            | 23.98             |
| 5     | 1.0                            | 15, 30 | 3 | 1.00              | 1.25              | 24.44             |
| 6     | 0.5                            | 15, 30 | 3 | 2.00              | 5.00              | 23.25             |
| 7     | 1.0                            | 15, 30 | 3 | 2.00              | 2.50              | 23.875            |
| 8     | 2.0                            | 15, 30 | 3 | 2.00              | 1.25              | 24.19             |

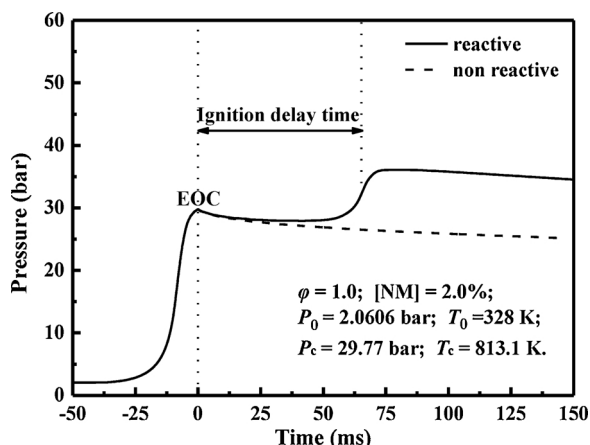


Fig. 2. Typical pressure trace and definition of ignition delay time.

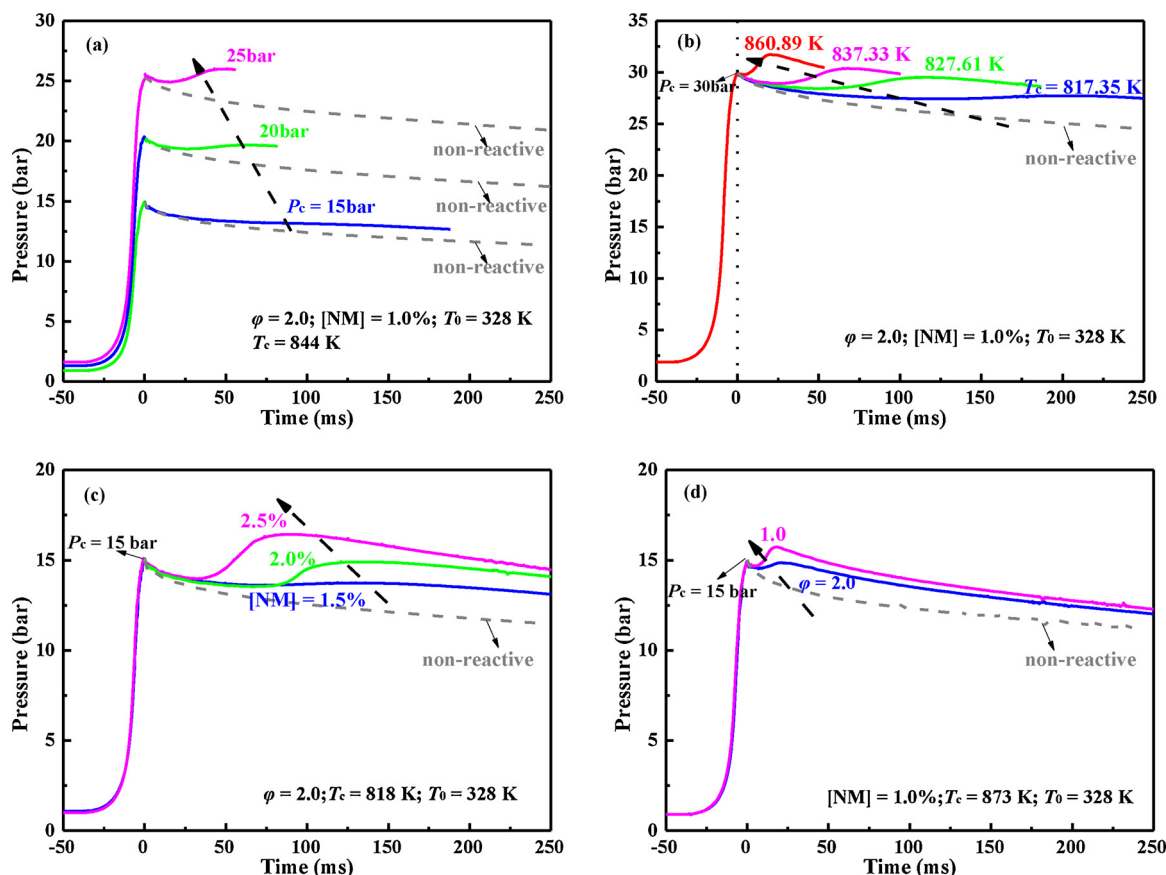


Fig. 3. Raw pressure histories at: (a) different EOC pressures; (b) different EOC temperatures; (c) different NM concentrations; (d) different equivalence ratios.

the time from the EOC to the onset of ignition, which is recognized as the instant with the maximum pressure rise rate. The  $P_c$  and  $T_c$  at EOC are used as the reference conditions for reporting ignition delay times [30,31].  $T_c$  is determined by  $P_c$  through the isentropic relation.

$$\int_{T_0}^{T_c} \frac{\gamma}{\gamma - 1} \frac{dT}{T} = \ln\left(\frac{P_c}{P_0}\right) \quad (1)$$

where  $P_c$  is the measured EOC pressure,  $T_c$  is the temperature at that time,  $T_0$  and  $P_0$  are the temperature and pressure before compression, and  $\gamma$  is the temperature dependent specific heat ratio.

The uncertainty of ignition delay measurement is estimated to be less than 12%, using the method reported in Refs. [32,33]. The uncertainty in ignition delay time is primarily resulted from the pressure measurement (including initial pressure by the pressure gauge, the dynamic pressure by the pressure transducer and charge amplifier) and the uncertainties in temperature measurement (dynamic pressure and the initial temperature). Uncertainty in the mixture composition is generally very small and is neglected. In this work, to avoid accidental errors, each experiment is repeated at least two times. By using the independent parameters methodology [34], the uncertainty in  $T_c$  is estimated to be less than 5 K. All the experimental data and the detailed uncertainty analysis can be seen in the Supplementary material.

Numerical simulations for ignition delay times were conducted using the zero-dimensional closed homogeneous reactor model in CHEMKIN-PRO. The volume-time profiles are adopted to account for the heat loss, while applying the zero-dimensional adiabatic core assumption. The ignition delay time in simulation is defined as the time interval between the start of calculation and the maximum pressure rise rate on the simulated pressure history.

### 3. Results and discussion

#### 3.1. Critical thermodynamic conditions for ignition

Fig. 3 shows the typical raw pressure histories of reactive (solid line) and non-reactive (dash line) tests at different EOC conditions. It can be seen that when nitromethane concentration is low, there will be an interesting slow pressure rise ratio phenomenon after EOC, which differs from common ignition. Fig. 3 (a) presents pressure-time histories with different EOC pressures. The pressure rise rate will increase and ignition delay time will decrease with the EOC pressure increase. Compared with the non-reactive pressure profile at EOC pressure of 15 bar, there is a weak ignition. That may be attributed to the low specific energy of nitromethane (about 11.3 MJ/kg) [35]. Fig. 3 (b) presents the evolution of pressure with different EOC temperatures. With the EOC temperature increasing, the ignition delay time will decrease. In addition, the nitromethane mixture goes through the process which is from non-ignition to weak ignition to ignition eventually with the temperature increase. The variation of pressure histories with the NM concentration is shown in Fig. 3(c). When the nitromethane concentration increases from 1.5% to 2.5%, the ignition explosion pressure and pressure rise ratio increase significantly and the ignition delay time decreases. Fig. 3(d) presents the evolution of pressure at different equivalence ratios. The ignition delay time decreases slightly with the equivalence ratio decreasing from 2.0 to 1.0. It illuminates that the reactivity of nitromethane is not sensitive to the amount of oxygen.

According to the raw pressure traces of nitromethane at low concentrations, three distinct regions were identified: 1. Ignition region. In this region, the ignition was defined as an obvious pressure rise after compression; 2. Weak ignition region. In this area, the rise of pressure is not distinct after the compression end point. However, the trace of pressure is higher than non-reactive history. That is to say there is a slow chemical reaction in this region; 3. Non-ignition region. Therefore, an ignition region diagram is presented for NM concentration of 1%, equivalence ratio of 2.0 and different EOC pressure, as shown in Fig. 4. It can be seen that the critical temperature of ignition increases with the EOC pressure decreasing, such as that at  $P_c = 15$  bar (about 850 K) is higher than that at  $P_c = 30$  bar (about 820 K). Increasing pressure is conducive to the occurrence of ignition. In this condition, the weak ignition region exists at some specific temperatures and pressures. As the pressure in EOC is increased to 30 bar, the weak ignition region becomes narrow. Because when pressure increases, the collision frequency of mixture gas molecules increases. Then chemical reaction will be accelerated. The weak ignition region will disappear as the pressure continues to increase. While, when the pressure is very low, high

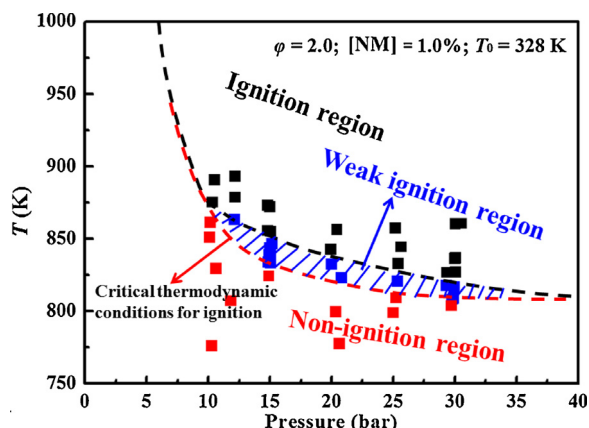


Fig. 4. Test points and regions about ignition at different temperatures (in EOC),  $P_c = 10$ –30 bar. (black symbols: ignition; blue symbols: weak ignition; red symbols: non-ignition). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

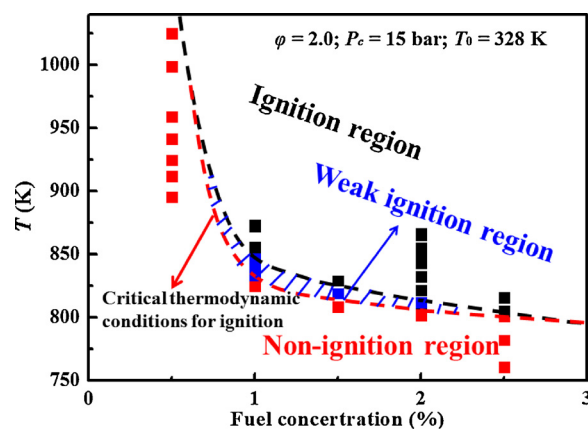


Fig. 5. Test points and regions about ignition at different temperatures (in EOC),  $P_c = 15$  bar,  $[NM] = 0.5\%$ –2.5%. (black symbols: ignition; blue symbols: weak ignition; red symbols: non-ignition). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

temperature will make the nitromethane mixture ignition, and there is no weak ignition region. This is because the high temperature will increase the internal energy of the mixture, which favors the reactivity such that the weak ignition region will also vanishes at high temperature.

Fig. 5 shows test points and predicted regions about ignition at different temperatures, different nitromethane concentrations and pressure of 15 bar. First, the critical ignition temperature of nitromethane mixture decreases with the concentration growing from 0.5% to 2.5%. Due to low specific energy, the ignition of nitromethane will have weak heat release at low NM concentrations due to slow chemical reaction. Weak ignition region will appear at low NM concentrations. As the NM concentration further grows, the weak ignition region becomes narrow. When the heat release can maintain combustion, the weak ignition region will disappear. In addition, when the NM concentration is less than 1%, it can be predicted that there will only be ignition region and non-ignition region. Since concentration is very low, there are no slow chemical reactions. Only when high temperature at EOC provide enough energy for mixture, it will ignition at the same time. Also there will be a critical NM concentration at which the ignition can't happen even at very high temperature. And for the NM concentration of 0.5%, there is no ignition at the temperature range from 895 to 1025 K.

#### 3.2. Ignition delay times: measurements and correlations

For the test experimental conditions in the ignition region, ignition delay time can be obtained, as in Fig. 2. Figs. 6–8 present all the measured ignition delay times as a function of inverse temperature for different equivalence ratios, NM concentrations and pressures. Across all test conditions in this work, it can be seen that the negative temperature coefficient behavior, which is typically observed for hydrocarbon fuels such as n-heptane and iso-octane in the low to intermediate temperature range, is absent for nitromethane. The measured IDTs in logarithm increase quasi-linearly with the inverse temperature, indicating its Arrhenius temperature dependence. These ignition delay times ( $\tau_{ign}$ , in ms) data are then correlated as a function of the experimental conditions, including the pressure ( $p$ , in bar), NM mole fraction ( $[NM]$ ), equivalence ratio ( $\phi$ ) and temperature ( $T$ , in K) through,

$$\tau_{ign} = Ap^B [NM]^C \phi^D \exp(E_a/RT) \quad (2)$$

where  $R = 1.986 \times 10^{-3}$  kcal/(mole·K) is the universal gas constant, and  $E_a$  is the global activation energy in kcal/mol. The fitting result is shown in Eq. (3), with  $R^2$  values above 0.97. The correlation of Eq. (3)



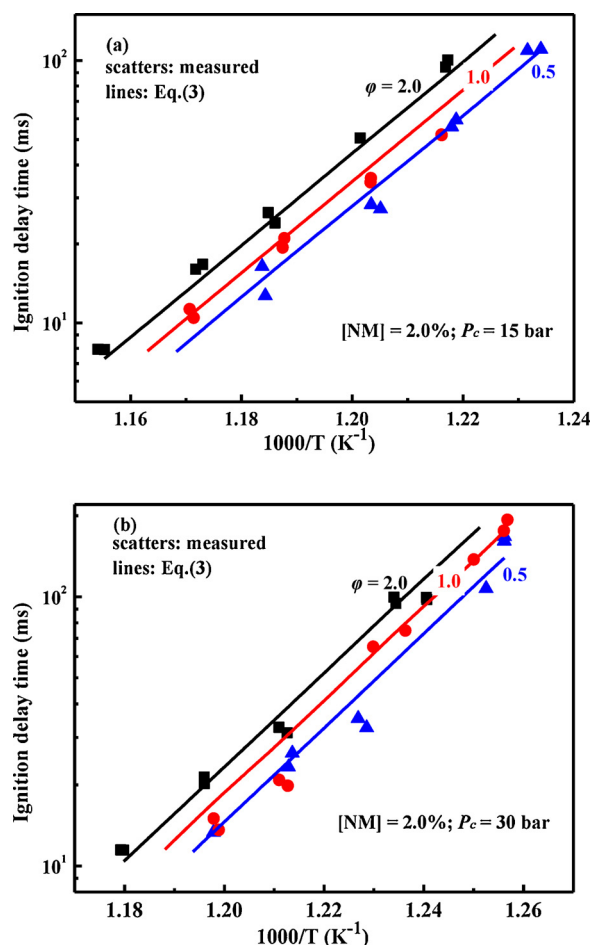


Fig. 6. Ignition delay times for [NM] = 2.0%,  $\phi = 0.5$ –2.0. (a)  $P_c = 15$  bar, (b)  $P_c = 30$  bar.

was also presented in Fig. 6–8. Though the ignition delay times at  $\phi = 1.0$  and NM concentration of 2.0% are slightly over-predicted, the correlation can give general good agreement against experimental results both quantitatively and qualitatively.

$$\tau_{\text{ign}} = 10^{-21.665} p^{-0.916} [\text{NM}]^{-1.963} \phi^{0.334} \exp(79.821/RT) \quad (3)$$

### 3.2.1. The effect of equivalence ratio

For the NM concentration of 2.0%, the effect of equivalence ratio on the ignition delay times of nitromethane at 15 bar and 30 bar is respectively illustrated in Fig. 6(a) and (b). It can be seen that the ignition delay times increase with increasing equivalence ratio at these two pressures. When the equivalence ratio increases with fixed NM concentration, the oxygen fraction decreases and the inert has a higher concentration, resulting in a higher ignition delay times. Previously for ignition delay times study in different shock tubes, similar effect of the equivalence ratio was observed [25,27].

### 3.2.2. The effect of NM concentration and pressure

The influence of NM concentration and the pressure on the ignition delay times of nitromethane was also presented in Figs. 7 and 8. The effect of NM concentration and pressure are very straightforward because the increase of the NM concentration and pressure increases the reactant NM concentration. For fixed pressure shown in Fig. 7, the ignition delay times decrease sharply with the NM concentration increasing. With the increase of the NM concentration, the oxygen concentration also increases for fixed equivalence ratio, faster reaction rate (assume global reaction  $k_{\text{gl}} [\text{NM}]^m [\text{O}_2]^n$ , where  $k_{\text{gl}}$  is the temperature

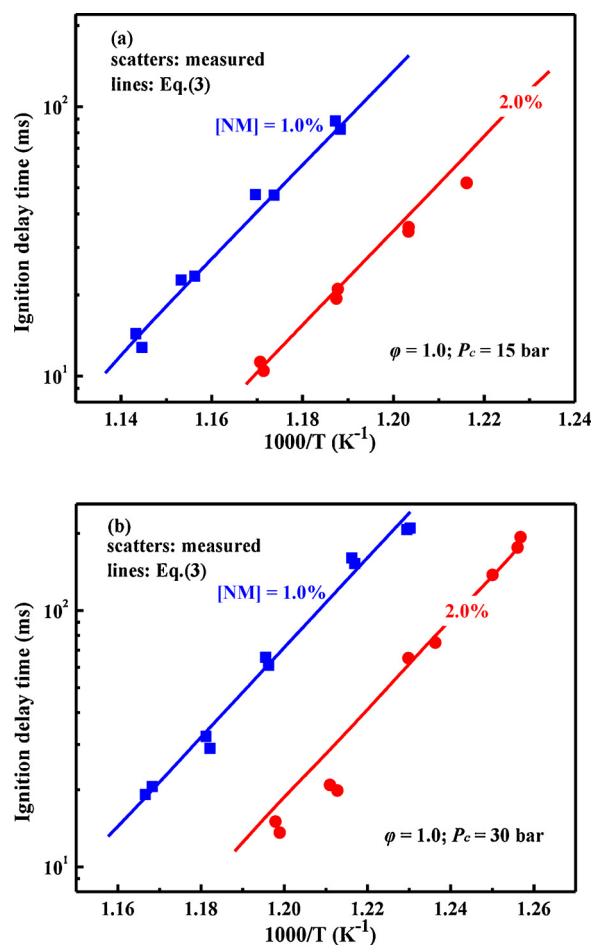


Fig. 7. Ignition delay times for  $\phi = 1.0$ , [NM] = 1.0% and 2.0%. (a)  $P_c = 15$  bar, (b)  $P_c = 30$  bar.

dependent rate constant,  $m$  and  $n$  is reaction order) is then expected due to increased.

Fig. 8 (a) shows experimental ignition delay times and fitted lines with Eq. (3) at equivalence ratio of 0.5 and pressure of 15 and 30 bar. It can be seen that the increase in pressure induces increased reactivity. Fig. 8 (b)–(c) presents the same trend under equivalence of 1.0 and 2.0. This is because higher pressure leads to higher reaction rate due to increased reactant concentration.

### 3.3. Comparisons of different kinetic mechanisms

#### 3.3.1. Low temperature ignition delay times in this work

The experimental results were compared with calculations using the kinetic mechanisms developed by Brequigny et al. [36], Mathieu et al. [25] and our previously developed model containing low temperature chemistry [27]. The Brequigny model, involving 88 species and 701 reactions, has been validated against laminar flame speeds, species profiles at low pressure and high temperature shock tube ignition delay times. The Mathieu model, involving 166 species and 1204 reactions, has been validated against high temperature ignition delay times, pyrolysis species profiles and laminar flame speeds. Our mechanism, involving 171 species and 1219 reactions, has been validated against high temperature ignition delay and laminar flame speeds. In addition, low temperature reactions have been updated including such as  $\text{CH}_3\text{NO}_2 + \text{CH}_3 = \text{CH}_2\text{NO}_2 + \text{CH}_4$ ,  $\text{CH}_3\text{NO}_2 + \text{OH} = \text{CH}_2\text{NO}_2 + \text{H}_2\text{O}$  and  $\text{CH}_3\text{NO}_2 + \text{H} = \text{CH}_2\text{NO}_2 + \text{H}_2$ . It is noted that all three mechanisms included the oxidation sub-model of nitromethane. But the models of Gao et al. as well as the models from Mathieu et al. and Brequigny

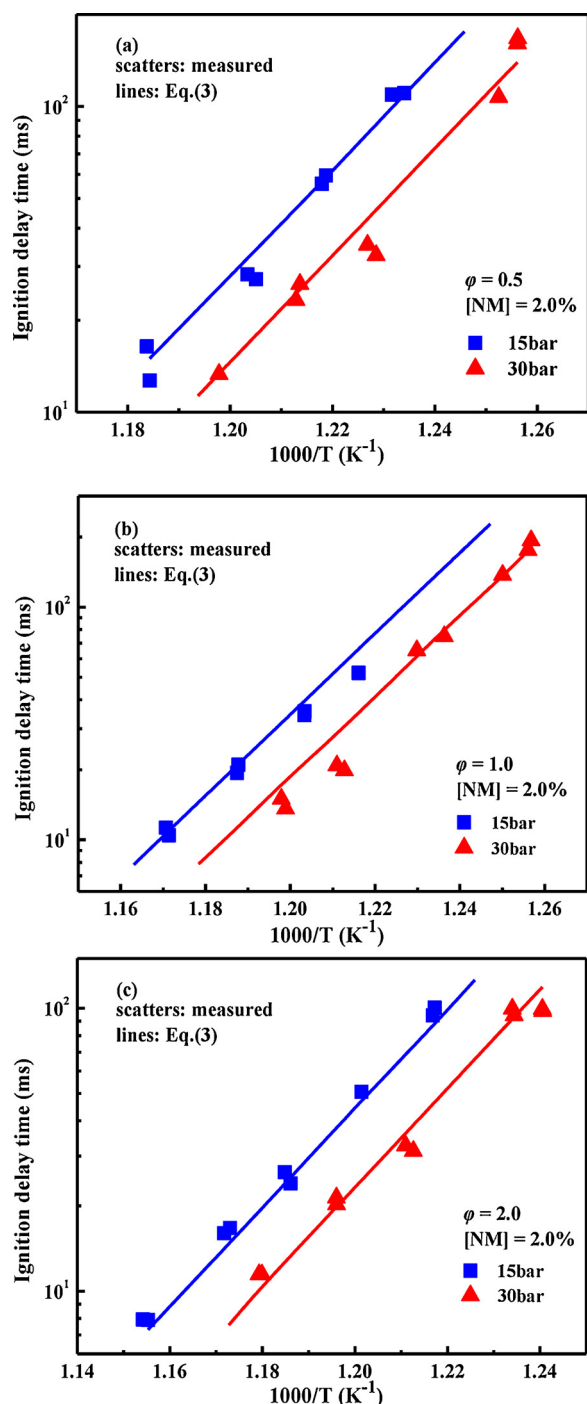


Fig. 8. Ignition delay times for  $[NM] = 2.0\%$  and  $P_c = 15$  bar and 30 bar. (a)  $\phi = 0.5$ , (b)  $\phi = 1.0$  and (c)  $\phi = 2.0$ .

et al. have not been sufficiently validated against low to intermediate temperature ignition delay of nitromethane.

Fig. 9 (a) presents the measured and calculated stoichiometric mixture ignition delay times of nitromethane with NM concentration of 1% at 15 bar and 30 bar. Generally, all three models can predict the trend of ignition delay times with temperature variation. The Brequigny model gives an over-prediction (less than 50% for all two pressures). At 15 bar, the predictions of Mathieu model have a good agreement with measured ignition delay times. At pressure of 30 bar, however, Mathieu model substantially underestimates the ignition delay times. The Gao model gives generally better predictions for ignition delay times at the two pressures, though at relatively lower temperature, predictions of

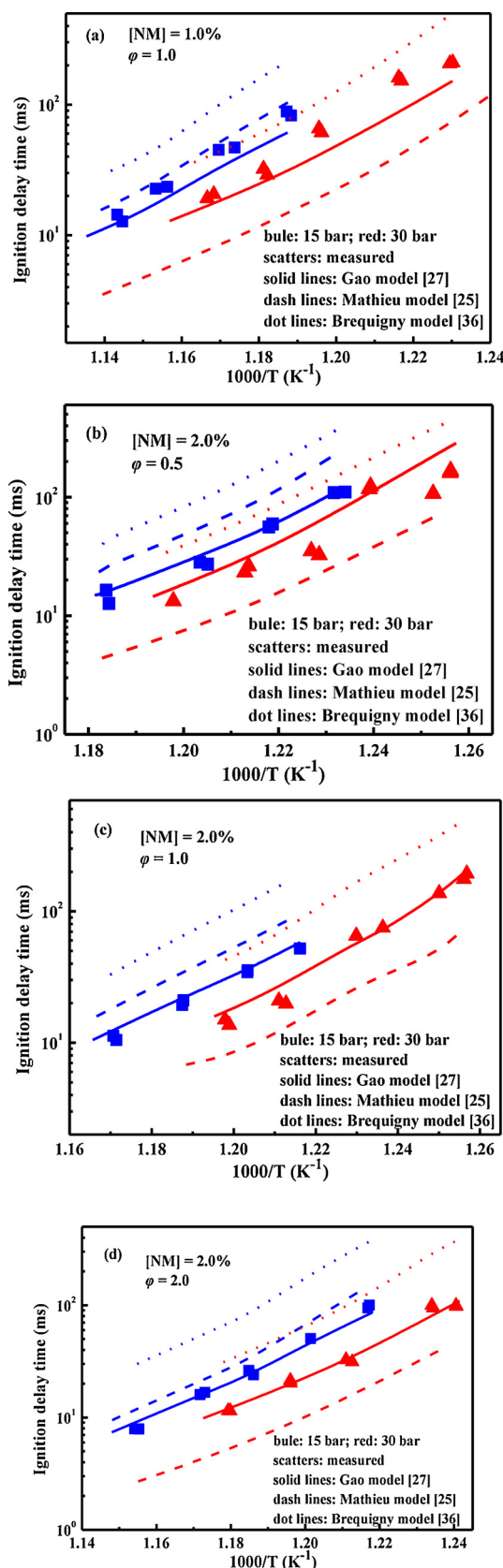


Fig. 9. Experimental and computed ignition delay times of nitromethane: (a)  $\phi = 1.0$ ,  $[NM] = 1.0\%$ ; (b)  $\phi = 0.5$ ,  $[NM] = 2.0\%$ ; (c)  $\phi = 1.0$ ,  $[NM] = 2.0\%$ ; (d)  $\phi = 2.0$ ,  $[NM] = 2.0\%$ .

the ignition delay times are slightly lower than measurements. We note that in Section 3.1, there is a narrow weak ignition region at NM concentration of 1% at relatively lower temperature. The pressure rise is slow, which affects the recognition of ignition onset and results in a larger ignition delay time. So when NM concentration is less than 1%, the measurements of ignition delay time are not used for comparison with the mechanism predictions.

Fig. 9(b)–(d) compares the measured and predicted ignition delay times of nitromethane at higher NM concentration (2%), different pressures and equivalence ratios. For this NM concentration mixture, there is almost no weak ignition region, as shown in Fig. 5. Therefore, the measured ignition delay times can be more accurately determined. In Fig. 9(b)–(d), similar to the trend observed in Fig. 9(a), the predictions using Brequigny model and Mathieu model show noticeable disagreement with the experimental data. Gao model again shows good performance in predicting the experimental ignition delay times.

### 3.3.2. CO and NO mole fraction evolution in literature

To further compare and validate the three mechanisms of nitromethane, the time evolution of NO and CO concentrations during the pyrolysis of the diluted nitromethane in a shock-tube is simulated and compared with the data collected by Hsu and Lin [21]. For the 0.25%  $\text{CH}_3\text{NO}_2/\text{Ar}$  mixture as shown in Fig. 10 (a), the Brequigny model [36] and Mathieu model [25] under-predict (more than 50%) noticeably the NO and CO evolution during pyrolysis. In comparison, the Gao model [27] predicts both NO and CO profiles with better agreements.

When nitromethane concentration is 0.4%, there is a reasonable prediction from three mechanisms as shown in Fig. 10 (b). In detail, the CO profile is well captured by the Gao model, whereas the Mathieu model slightly under-predicts the CO formation. The equilibrium NO is

over-estimated by all the models by 15%.

For 0.76%  $\text{CH}_3\text{NO}_2$ , the NO profile is relatively well predicted, and CO profile is under-predicted (around 30%) by the Brequigny model and Mathieu model, while the Gao model over-predicts NO profile by up to 20% but predicts CO profile well, as shown in Fig. 10(c).

We note that the Brequigny model and Mathieu model can't repeat the ignition delay time data well at low to intermediate temperature and the NO and CO concentration profiles at relatively lower temperature. While the Gao model shows good performance in predicting the ignition delay times and reasonable results in predicting the NO and CO concentration profiles. To further understand Gao model at low to intermediate temperature chemistry, chemical kinetic analysis has been conducted in the following.

## 3.4. Chemical kinetic analysis

### 3.4.1. Sensitivity analysis

Sensitivity analysis was conducted to identify the controlling reactions in predicting the auto-ignition process.

There are hundreds of reactions in the Gao model [27], as mentioned previously. The sensitivity coefficient of the  $i^{\text{th}}$  reaction  $SC_i$ , is calculated through Eq. (4),

$$SC_i = (\tau_{2ki} - \tau_{0.5ki}) / \tau_{ki} \quad (4)$$

where  $k_i$  is the rate constant of the  $i^{\text{th}}$  reaction,  $\tau_{ki}$  is the correspondingly computed ignition delay times. A negative  $SC_i$  indicates that increasing the rate constant  $k_i$  reduces the ignition delay time  $\tau_{ki}$ , thus the  $i^{\text{th}}$  reaction plays a promoting role for ignition, and vice versa. As such, by calculating and comparing the sensitivity coefficients for all the reactions, we can find out the most sensitive reactions for the ignition delay

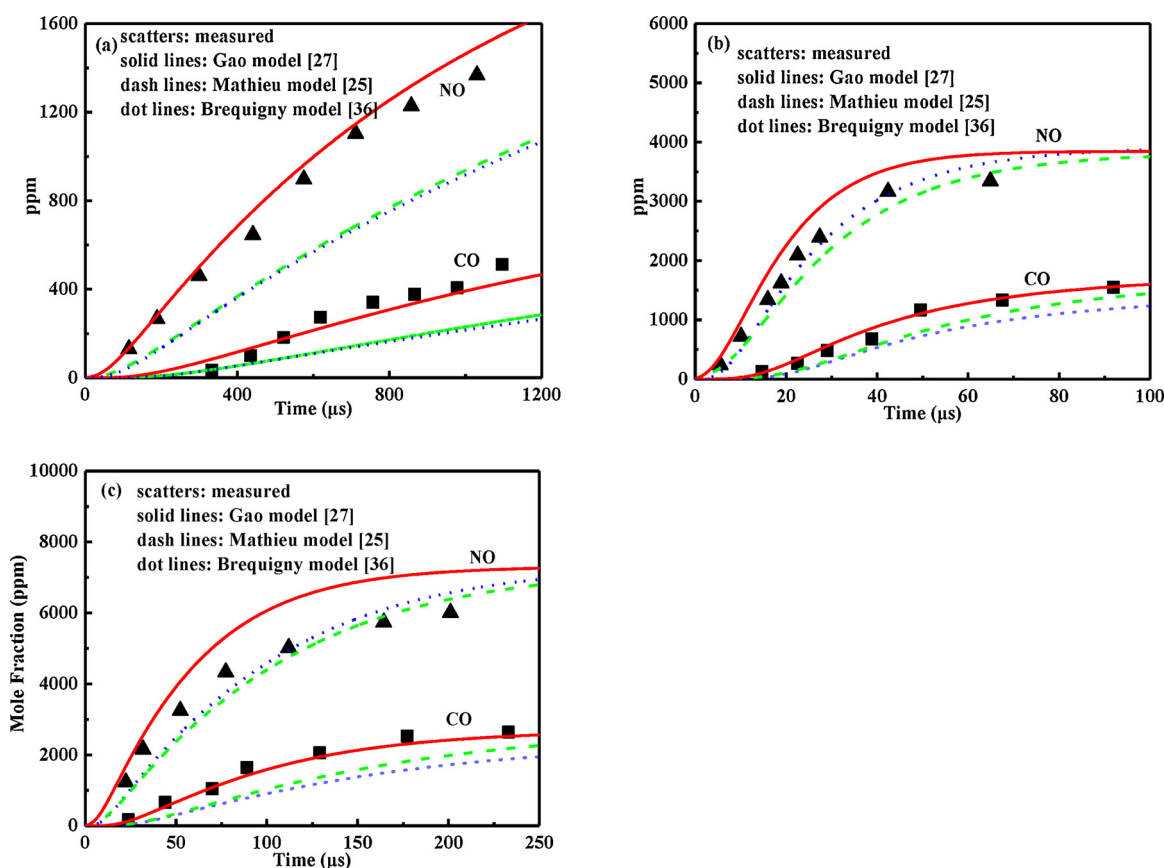


Fig. 10. Comparison of measured [21] and predicted concentrations of CO and NO in diluted shock tube decomposition of nitromethane. Experimental conditions: incident shock; (a): 0.25%  $\text{CH}_3\text{NO}_2/\text{Ar}$  mixture;  $T = 1051 \text{ K}$  (NO) or  $1057 \text{ K}$  (CO);  $P = 0.83 \text{ atm}$ ; (b): 0.40%  $\text{CH}_3\text{NO}_2/\text{Ar}$  mixture;  $T = 1392 \text{ K}$  (NO) or  $1380 \text{ K}$  (CO);  $P = 0.5 \text{ atm}$ ; (c): 0.76%  $\text{CH}_3\text{NO}_2/\text{Ar}$  mixture;  $T = 1243 \text{ K}$  (NO) or  $1236 \text{ K}$  (CO);  $P = 0.69 \text{ atm}$  (NO) or  $0.65 \text{ atm}$  (CO).





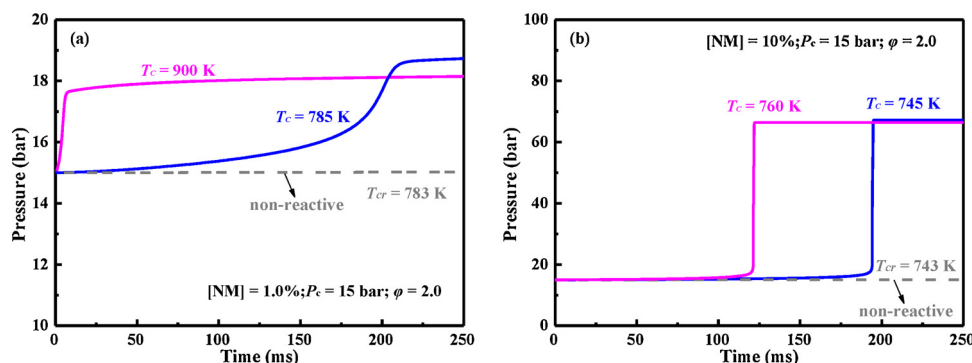


Fig. 13. Simulated typical pressure evolution: (a) the three types of ignition behavior at NM concentration of 1.0%; (b) ignition and non-ignition behavior at NM concentration of 10%.

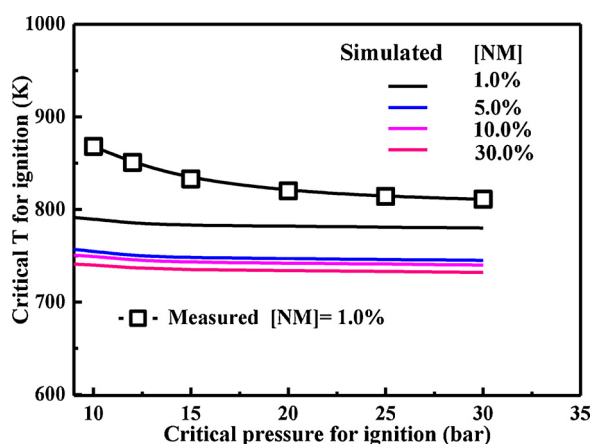


Fig. 14. Measured (scatters for 1.0% NM only) and predicted (lines) critical conditions for ignition for the equivalence ratio of 2.0.

conditions that are beyond the ability of the experimental facility were predicted.

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## Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.jhazmat.2019.05.036>.

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